

DAX Measures Reference Guide

Social Media Analytics Dashboard

Power BI · Excel · DAX · 2026

37 Total Measures	7 Categories	8 Core Metrics	6 KPI Comparisons
---	------------------------------------	--------------------------------------	---

How to Use This Guide

This reference documents every DAX measure behind the Social Media Analytics Dashboard. Each entry includes the full formula and a plain-English explanation of what it calculates and why.

Base measures first

Measures like Total Engagement, Total Impressions, and Total Cost are referenced throughout the model. Build these before anything else.

DIVIDE over division

All division uses `DIVIDE(numerator, denominator, 0)` — the third argument returns 0 instead of an error when the denominator is blank or zero.

VAR for readability

Complex measures use VAR to store intermediate results. This keeps formulas readable, debuggable, and avoids repeated calculations.

ALL() removes filters

To make a measure ignore slicers (e.g. global averages, viral thresholds), wrap it in `CALCULATE(..., ALL(table))`.

DATEADD for time intelligence

Month-over-month comparisons use `DATEADD('Calendar'[Date], -1, MONTH)` — this requires a properly marked Date table in the model.

TOPN + MAXX pattern

The dynamic insight measures use TOPN to find the top row of a virtual table, then MAXX to extract the value from that row.

Contents

37 measures across 7 categories.

01 — CORE METRICS 8 measures

Total Engagement — Sum of all engagement actions across all posts and platforms.

Total Impressions — Total number of times content was displayed to users.

Total Reach — Total unique accounts that saw the content.

Total Cost — Combined production and promotion spend across all posts.

Total Posts — Count of all posts in the current filter context.

Total Followers — Total followers gained across all platforms and campaigns.

Avg Reach — Average reach per post in the current filter context.

Avg Engagement — Average engagement per post. Used as the baseline for the Viral Flag measure.

02 — RATE & EFFICIENCY 7 measures

Engagement Rate — Percentage of impressions that resulted in an engagement action.

Likes % — Likes as a share of total engagements.

Comment % — Comments as a share of total engagements.

Shares % — Shares as a share of total engagements.

Saves % — Saves as a share of total engagements.

CTR (Click-Through Rate) — Link clicks divided by total impressions.

Follower Conversion Rate — What percentage of reached users converted to followers.

03 — COST & ROI MEASURES 6 measures

Cost Per Engagement — Average cost for every single engagement action.

CPM (Cost Per Mille) — Cost per 1,000 impressions. Industry-standard ad efficiency metric.

Cost Per Follower — Average cost to acquire one new follower.

ROI Engagement — Engagement return per unit of cost — higher is better.

Organic CPE — Cost per engagement for non-boosted (organic) posts only.

Boosted CPE — Cost per engagement for boosted (paid) posts only.

04 — ADVANCED SCORING 3 measures

Engagement Quality Score — Weighted engagement score. Comments (3x), Saves (4x), and Shares (5x) carry more weight than Likes (1x), reflecting real audience intent.

Campaign Efficiency Index — Composite efficiency score combining engagement and follower growth relative to total spend. Followers are weighted 2x to reflect their long-term value.

Avg Engagement Rate (Global) — Engagement rate calculated across all data, ignoring the current filter context. Used as the reference line on the Engagement Trend chart.

05 — KPI COMPARISON (MONTH-OVER-MONTH) 6 measures

Engagement MoM KPI — Returns dynamic month-over-month text, e.g. "vs Jul-25 ▲ 7.2%". Returns blank when no single month is selected.

Impressions KPI — Same month-over-month pattern applied to Total Impressions.

Reach KPI — Same month-over-month pattern applied to Total Reach.

Engagement KPI — Same month-over-month pattern applied to Total Engagement.

CPE KPI — Same month-over-month pattern applied to Cost Per Engagement.

ROI KPI — Same month-over-month pattern applied to ROI Engagement.

06 — DYNAMIC INSIGHT TEXT 4 measures

Viral Flag — Labels a post as "Viral" if its total engagement exceeds five times the global average. Uses ALL() to ensure the comparison is always against unfiltered data.

Monthly Engagement Summary — Generates the engagement trend paragraph below the monthly chart. Identifies the best and worst months and the top platform automatically.

Platform Media Insight — Generates insight text for the Platform-Content matrix. Identifies the top-performing platform and content type dynamically.

Strategic Summary — Synthesises platform, content type, campaign, and time slot into a single dynamic paragraph for the Strategic Insights section.

07 — ROW-LEVEL HELPER MEASURES 3 measures

Post Platform — Row-level helper that returns the platform for a single post. Used in the Top Posts table.

Media Type — Row-level helper that returns the content type for a single post.

Boosted — Row-level helper that returns whether a post was boosted. Used in the Top Posts table.

Fundamental aggregation measures that form the foundation of the dashboard. These base measures are referenced throughout every other category.

Total Engagement

Sum of all engagement actions across all posts and platforms.

```
=SUM(social_media_data[Engagements])
```

Total Impressions

Total number of times content was displayed to users.

```
=SUM(social_media_data[Impressions])
```

Total Reach

Total unique accounts that saw the content.

```
=SUM(social_media_data[Reach])
```

Total Cost

Combined production and promotion spend across all posts.

```
=SUM(social_media_data[Production_Cost])  
+ SUM(social_media_data[Promotion_Budget])
```

Total Posts

Count of all posts in the current filter context.

```
=COUNT(social_media_data[Post_ID])
```

Total Followers

Total followers gained across all platforms and campaigns.

```
=SUM(social_media_data[Followers_Gained])
```

Avg Reach

Average reach per post in the current filter context.

```
=AVERAGE(social_media_data[Reach])
```

Avg Engagement

Average engagement per post. Used as the baseline for the Viral Flag measure.

```
=DIVIDE(  
    [Total Engagement],  
    COUNTROWS(social_media_data)  
)
```

02 Rate & Efficiency

7 measures

Calculated rates and efficiency scores that express performance relative to spend, reach, or impressions.

Engagement Rate

Percentage of impressions that resulted in an engagement action.

```
=DIVIDE([Total Engagement], [Total Impressions], 0)
```

Likes %

Likes as a share of total engagements.

```
=DIVIDE(  
    SUM(social_media_data[Likes]),  
    [Total Engagement],  
    0  
)
```

Comment %

Comments as a share of total engagements.

```
=DIVIDE(  
    SUM(social_media_data[Comments]),  
    [Total Engagement],  
    0  
)
```

Shares %

Shares as a share of total engagements.

```
=DIVIDE(  
    SUM(social_media_data[Shares]),  
    [Total Engagement],  
    0  
)
```

Saves %

Saves as a share of total engagements.

```
=DIVIDE(  
    SUM(social_media_data[Saves]),  
    [Total Engagement],  
    0  
)
```

CTR (Click-Through Rate)

Link clicks divided by total impressions.

```
=DIVIDE(  
    SUM(social_media_data[Link_Clicks]),  
    SUM(social_media_data[Impressions]),  
    0  
)
```

Follower Conversion Rate

What percentage of reached users converted to followers.

```
=DIVIDE(  
    SUM(social_media_data[Followers_Gained]),  
    SUM(social_media_data[Reach]),  
    0  
)
```

Cost efficiency measures that evaluate the financial performance of campaigns and content strategies.

Cost Per Engagement

Average cost for every single engagement action.

```
=DIVIDE(  
    [Total Cost],  
    [Total Engagement],  
    0  
)
```

CPM (Cost Per Mille)

Cost per 1,000 impressions. Industry-standard ad efficiency metric.

```
=DIVIDE(  
    [Total Cost],  
    SUM(social_media_data[Impressions]),  
    0  
) * 1000
```

Cost Per Follower

Average cost to acquire one new follower.

```
=DIVIDE(  
    [Total Cost],  
    SUM(social_media_data[Followers_Gained]),  
    0  
)
```

ROI Engagement

Engagement return per unit of cost — higher is better.

```
=DIVIDE([Total Engagement], [Total Cost], 0)
```

Organic CPE

Cost per engagement for non-boosted (organic) posts only.

```
=CALCULATE(  
    [Cost Per Engagement],  
    social_media_data[Is_Boosted] = "No"  
)
```

Boosted CPE

Cost per engagement for boosted (paid) posts only.

```
=CALCULATE(  
    [Cost Per Engagement],  
    social_media_data[Is_Boosted] = "Yes"  
)
```

Composite scores that weight different engagement types and combine multiple metrics into a single performance index.

Engagement Quality Score

Weighted engagement score. Comments (3x), Saves (4x), and Shares (5x) carry more weight than Likes (1x), reflecting real audience intent.

```
=(SUM(social_media_data[Likes]) * 1) +  
  (SUM(social_media_data[Comments]) * 3) +  
  (SUM(social_media_data[Shares]) * 5) +  
  (SUM(social_media_data[Saves]) * 4)
```

Campaign Efficiency Index

Composite efficiency score combining engagement and follower growth relative to total spend. Followers are weighted 2x to reflect their long-term value.

```
=DIVIDE(  
    [Total Engagement]  
    + (SUM(social_media_data[Followers_Gained]) * 2),  
    [Total Cost],  
    0  
)
```

Avg Engagement Rate (Global)

Engagement rate calculated across all data, ignoring the current filter context. Used as the reference line on the Engagement Trend chart.

```
=CALCULATE(  
    [Engagement Rate],  
    ALL(social_media_data)  
)
```

Dynamic text measures that display month-over-month change with directional arrows. Each follows the same VAR pattern: current value, prior value, percent change, then a conditional RETURN.

Engagement MoM KPI

Returns dynamic month-over-month text, e.g. "vs Jul-25 ▲ 7.2%". Returns blank when no single month is selected.

```
=VAR Curr = [Engagement Rate]
VAR PrevRate =
    CALCULATE(
        [Engagement Rate],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR PrevMonthDate =
    CALCULATE(
        MAX('Calendar'[Date]),
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR PrevMonthName = FORMAT(PrevMonthDate, "MMM-YY")
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                IF(ChangePct < 0,
                    "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%"),
                    "vs " & PrevMonthName & " ■ 0%"
                )
            )
        )
    )
```

Impressions KPI

Same month-over-month pattern applied to Total Impressions.

```
=VAR Curr = [Total Impressions]
VAR PrevRate =
    CALCULATE(
        [Total Impressions],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
VAR PrevMonthName =
    FORMAT(
        CALCULATE(
            MAX('Calendar'[Date]),
            DATEADD('Calendar'[Date], -1, MONTH)
        ),
        "MMM-YY"
    )
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%")
            )
        )
    )
```

Reach KPI

Same month-over-month pattern applied to Total Reach.

```
=VAR Curr = [Total Reach]
VAR PrevRate =
    CALCULATE(
        [Total Reach],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
VAR PrevMonthName =
    FORMAT(
        CALCULATE(
            MAX('Calendar'[Date]),
            DATEADD('Calendar'[Date], -1, MONTH)
        ),
        "MMM-YY"
    )
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%")
            )
        )
    )
```

Engagement KPI

Same month-over-month pattern applied to Total Engagement.

```
=VAR Curr = [Total Engagement]
VAR PrevRate =
    CALCULATE(
        [Total Engagement],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
VAR PrevMonthName =
    FORMAT(
        CALCULATE(
            MAX('Calendar'[Date]),
            DATEADD('Calendar'[Date], -1, MONTH)
        ),
        "MMM-YY"
    )
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%")
            )
        )
    )
```

CPE KPI

Same month-over-month pattern applied to Cost Per Engagement.

```
=VAR Curr = [Cost Per Engagement]
VAR PrevRate =
    CALCULATE(
        [Cost Per Engagement],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
VAR PrevMonthName =
    FORMAT(
        CALCULATE(
            MAX('Calendar'[Date]),
            DATEADD('Calendar'[Date], -1, MONTH)
        ),
        "MMM-YY"
    )
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%")
            )
        )
    )
```

ROI KPI

Same month-over-month pattern applied to ROI Engagement.

```
=VAR Curr = [ROI Engagement]
VAR PrevRate =
    CALCULATE(
        [ROI Engagement],
        DATEADD('Calendar'[Date], -1, MONTH)
    )
VAR ChangePct = DIVIDE(Curr - PrevRate, PrevRate)
VAR PrevMonthName =
    FORMAT(
        CALCULATE(
            MAX('Calendar'[Date]),
            DATEADD('Calendar'[Date], -1, MONTH)
        ),
        "MMM-YY"
    )
RETURN
    IF(
        NOT HASONEVALUE('Calendar'[MMM-YYYY]), "",
        IF(ISBLANK(PrevRate), "No Prior Month",
            IF(ChangePct > 0,
                "vs " & PrevMonthName & " ▲ " & FORMAT(ChangePct, "0.0%"),
                "vs " & PrevMonthName & " ▼ " & FORMAT(ABS(ChangePct), "0.0%")
            )
        )
    )
```

Measures that use VAR chains, TOPN, and string concatenation to generate natural-language insight text that updates dynamically with every slicer selection.

Viral Flag

Labels a post as "Viral" if its total engagement exceeds five times the global average. Uses ALL() to ensure the comparison is always against unfiltered data.

```
=VAR GlobalAvg =  
    CALCULATE(  
        [Avg Engagement],  
        ALL(social_media_data)  
    )  
RETURN  
    IF(  
        [Total Engagement] > GlobalAvg * 5,  
        "Viral",  
        "Normal"  
    )
```

Monthly Engagement Summary

Generates the engagement trend paragraph below the monthly chart. Identifies the best and worst months and the top platform automatically.

```
=VAR TopMonthTable =
    TOPN(1, VALUES('Calendar'[Month]), [Engagement Rate], DESC)
VAR BottomMonthTable =
    TOPN(1, VALUES('Calendar'[Month]), [Engagement Rate], ASC)
VAR TopMonth = MAXX(TopMonthTable, 'Calendar'[Month])
VAR TopRate = MAXX(TopMonthTable, [Engagement Rate])
VAR BottomMonth = MAXX(BottomMonthTable, 'Calendar'[Month])
VAR BottomRate = MAXX(BottomMonthTable, [Engagement Rate])
VAR AvgRate =
    CALCULATE([Engagement Rate], ALL('Calendar'[Month]))
VAR TopPlatform =
    MAXX(
        TOPN(1,
            SUMMARIZE(
                social_media_data,
                social_media_data[Platform],
                "Eng", [Total Engagement]
            ),
            [Eng], DESC
        ),
        social_media_data[Platform]
    )
RETURN
    "Engagement peaked in " & TopMonth
    & " at " & FORMAT(TopRate, "0.00%")
    & ", driven primarily by " & TopPlatform
    & " content. The lowest engagement occurred in "
    & BottomMonth & " at " & FORMAT(BottomRate, "0.00%")
    & ". Average engagement across the period is "
    & FORMAT(AvgRate, "0.00%") & "."
```

Platform Media Insight

Generates insight text for the Platform-Content matrix. Identifies the top-performing platform and content type dynamically.

```
=VAR TopPlatformTable =  
    TOPN(1, VALUES(social_media_data[Platform]),  
        [Engagement Rate], DESC)  
VAR TopPlatform = MAXX(TopPlatformTable, social_media_data[Platform])  
VAR PlatformRate = MAXX(TopPlatformTable, [Engagement Rate])  
VAR TopContentTable =  
    TOPN(1, VALUES(social_media_data[Content_Type]),  
        [Engagement Rate], DESC)  
VAR TopContent = MAXX(TopContentTable, social_media_data[Content_Type])  
VAR ContentRate = MAXX(TopContentTable, [Engagement Rate])  
RETURN  
    TopPlatform  
    & " delivers the highest engagement among all platforms"  
    & " with an average engagement rate of "  
    & FORMAT(PlatformRate, "0.00%")  
    & ". Across media formats, " & TopContent  
    & " posts perform best with an engagement rate of "  
    & FORMAT(ContentRate, "0.00%") & "."
```

Strategic Summary

Synthesises platform, content type, campaign, and time slot into a single dynamic paragraph for the Strategic Insights section.

```
=VAR TopPlatform =  
    MAXX(  
        TOPN(1, VALUES(social_media_data[Platform]),  
            [Engagement Rate], DESC),  
        social_media_data[Platform]  
    )  
VAR PlatformRate =  
    MAXX(  
        TOPN(1, VALUES(social_media_data[Platform]),  
            [Engagement Rate], DESC),  
        [Engagement Rate]  
    )  
VAR TopContent =  
    MAXX(  
        TOPN(1, VALUES(social_media_data[Content_Type]),  
            [Engagement Rate], DESC),  
        social_media_data[Content_Type]  
    )  
VAR TopCampaign =  
    MAXX(  
        TOPN(1, VALUES(social_media_data[Campaign]),  
            [Total Engagement], DESC),  
        social_media_data[Campaign]  
    )  
VAR BestTime =  
    MAXX(  
        TOPN(1, VALUES(social_media_data[Post_Time_Slot]),  
            [Engagement Rate], DESC),  
        social_media_data[Post_Time_Slot]  
    )  
RETURN  
    TopPlatform  
    & " drives the highest engagement across platforms"  
    & " with an average rate of "  
    & FORMAT(PlatformRate, "0.0%") & ". "  
    & TopContent  
    & " content generates the strongest audience interaction,"  
    & " while the " & TopCampaign  
    & " campaign produced the highest total engagement."  
    & " Posting during the " & BestTime  
    & " period consistently delivers stronger engagement."
```

Lightweight helper measures used at the row level to support the Top Posts table and other detail-level visuals.

Post Platform

Row-level helper that returns the platform for a single post. Used in the Top Posts table.

```
=MAX(social_media_data[Platform])
```

Media Type

Row-level helper that returns the content type for a single post.

```
=MAX(social_media_data[Content_Type])
```

Boosted

Row-level helper that returns whether a post was boosted. Used in the Top Posts table.

```
=MAX(social_media_data[Is_Boosted])
```

This guide documents the complete DAX layer behind the Social Media Analytics Dashboard, covering aggregation, rate and cost efficiency, composite scoring, time-intelligence, and dynamic natural-language insight generation.